

KhataAI Software Design Document

Section 2 – System Architecture

Objective: Define the production-ready backend architecture before implementation.

2.1 High-Level Architecture

React Web → Nginx (HTTPS) → FastAPI → MongoDB Atlas.

Redis stores temporary data (OTP, rate limits, cache).

Background workers handle OTP SMS, PDF generation, WhatsApp messages and notifications.

External AI services: Speech-to-Text + Gemini.

Architecture Flow

React Client

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Nginx Reverse Proxy

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FastAPI Backend

- Authentication
- Customer Module
- Product Module
- Transactions
- Voice AI
- Analytics

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MongoDB + Redis + Background Tasks + AI Services

2.2 Layered Backend Architecture

Request → Router → Service → Repository → MongoDB → Repository → Service → Response.

Router: validation/authentication.

Service: business rules.

Repository: database only.

2.3 AI Architecture

Audio → SpeechService → Transcript → LLMService (Gemini) → InvoiceExtractionService → ProductMatchingService → Validation → Preview Screen → Confirmation → Save Transaction.

2.4 Authentication Architecture

Phone Number → Generate OTP → Store OTP in Redis → Background SMS → Verify OTP → Generate JWT Access & Refresh Tokens → Access Application.

2.5 Voice Billing Flow

Vendor speaks → Upload Audio → Speech-to-Text → Gemini Structured JSON → Product Matching → Backend GST Calculation → Invoice Draft → User Confirmation → Save →

Generate PDF.

2.6 Request Lifecycle

Client → JWT Middleware → Router → Pydantic Validation → Service → Repository → MongoDB → Response Model → JSON Response.

2.7 Background Tasks

Run asynchronously:

- Send OTP SMS
- Generate Invoice PDF
- Send WhatsApp Invoice
- Create Audit Log
- Low Stock Notification
- Daily Reports

2.8 Redis Usage

OTP Storage, Rate Limiting, Login Attempts, AI Cache, Product Search Cache, JWT Blacklist.

Do NOT store customers, products or transactions in Redis.

2.9 Security

HTTPS only, JWT Authentication, Role-Based Access Control, Multi-tenant isolation using store_id from JWT, bcrypt password hashing, audit logging.

2.10 AWS Deployment

CloudFront (future) → Nginx → EC2 (FastAPI) → MongoDB Atlas + Redis → S3 (PDFs) → CloudWatch Logs.

2.11 Dedicated AI Services

SpeechService - Speech-to-text provider.

LLMService - Gemini structured extraction.

ProductMatchingService - Alias & fuzzy matching.

InvoiceExtractionService - Validate extracted JSON and prepare invoice draft.

Architecture Decisions (Frozen)

Decision	Status
FastAPI Backend	Approved
MongoDB	Approved
Repository Pattern	Approved
Service Layer	Approved
OTP Authentication	Approved
JWT Access + Refresh	Approved
Redis for Temporary Data	Approved

Preview Before Save	Approved
Immutable Transactions	Approved
Multi-Tenant Isolation	Approved
Dedicated AI Services	Approved

Notes for Implementation

This architecture is intentionally modular so that each component can be tested independently. No business logic should exist in routers. AI never calculates financial values; it only extracts structured data. The backend remains the single source of truth for GST, pricing, inventory and ledger calculations.